



Posterior Crown Preparations

Joel Whiteman, DDS

7/13/2026

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Learning Outcomes & Competency Statements

Learning Outcomes

1. Understand the process of treatment planning for a crown.
2. Understand the steps of posterior crown preparation.

Competency statements met #1.4, 4.1, 5.1

The Learning Outcomes for this lecture are to understand the process for treatment planning for a crown, be able to apply the knowledge of preparation principles to design a crown preparation and understand the steps of posterior crown preparation.

Treatment Planning

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What is treatment planning, it is a logical sequence of treatment designed to restore the patient's dentition to good health, optimal function and appearance. When we are treatment planning it is essential that we take a patient's preferences and needs into account. They may not want our ideal plan for them. When we are planning for a crown, it is necessary for the treatment to achieve one or more of the following conditions.

- 1) Correct Existing Disease
- 2) Prevent Future Disease
- 3) Restore Function
- 4) Improve Appearance

Sometimes this can be achieved with a partial coverage restoration as we have discussed earlier, but this relies on the residual tooth having the strength and integrity to be stable under function. At times we will need to reinforce the tooth with a full coverage restoration. Also, that full coverage will help us re-establish a more ideal occlusion.

Traditionally, if a filling extended more than 1/3 of the occlusal table a crown was recommended. With modern porcelain restorations this is not always the case. You will need to look at the size of the existing restoration but also how much residual tooth structure will be left.

Treatment Planning



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The pictures in this slide represent common situations that you will see in clinical practice. The left side picture has a large DOL amalgam that has the distal buccal cusp broken off. The residual tooth structure remaining after you remove the old filling will be minimal. This is not an ideal situation for an onlay. The best course of action would be to place a build-up and do a full coverage crown. The righthand picture is a similar situation. Since this amalgam occupies most of the occlusal table all of the cusps will need to be covered. This coverage can be done with a large onlay, crownlay or full coverage crown. Realistically, many insurance companies do not cover a partial coverage crown so patients will select a full coverage crown as the treatment they want. Also, in teeth that have been severely weakened by prior restorations an intracoronal restoration will generate a greater wedging effect. The weakened tooth structure will not be able to resist these forces and will break.

Full Coverage Crown



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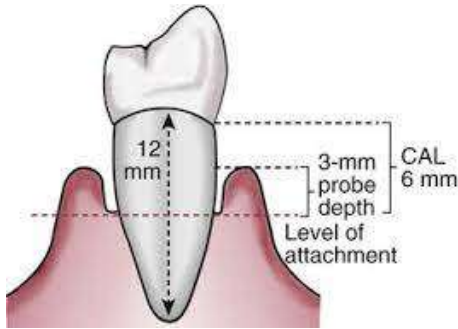
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A full coverage crown, also known as a cap, is an extracoronal restoration. It encircles all or part of the remaining tooth structure and occlusal surface. Crowns can strengthen teeth that have been weakened by caries or trauma. When we prepare a full coverage restoration, we have to remove a lot of tooth structure to obtain proper material thickness to resist breakage. Lastly, the margin will be near or below the free gingival margin which can make it challenging for the patient to maintain gingival health in that area.

A full coverage crown will need to fit well to the tooth to minimize the tooth exposure of the luting agent, or cement, to oral fluids. This will increase the longevity of the restoration. The internal dimensions of the tooth need to be designed to allow the restoration to seat without binding against the vertical prepared walls while remaining stable and not becoming displaced during functioning.

Periodontal Considerations



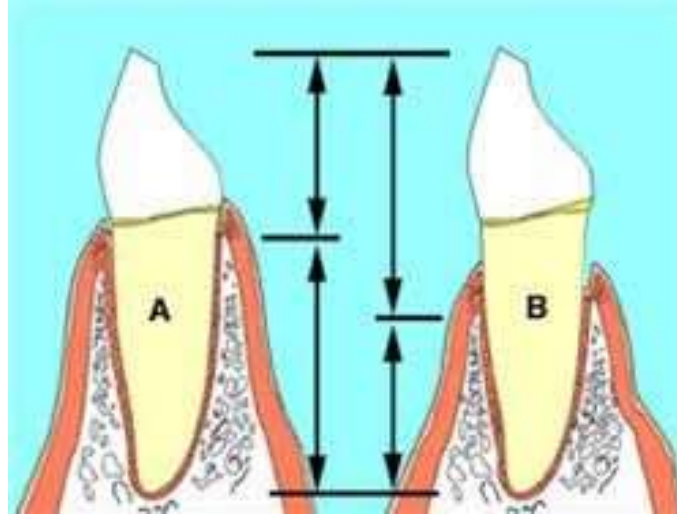
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When you are planning for a crown, it is important to consider the periodontal condition of the tooth. As you know the periodontium forms the foundation for the teeth. If this is weakened or compromised, it will not matter how amazing your crown is the tooth will fail. We need to pay attention to two factors, they are attachment loss and furcation exposure. The prognosis of a tooth dramatically decreases with greater attachment loss and greater furcation exposure. A tooth with 50% attachment loss and a Class 3 furcation would have a poor overall prognosis. Placing a crown on this type of tooth would be questionable because this type of tooth might not last that long in the patient's mouth. A way we dentist evaluate this is by looking at the crown to root ratio. We will discuss this on the next slide.

Crown to Root Ratio



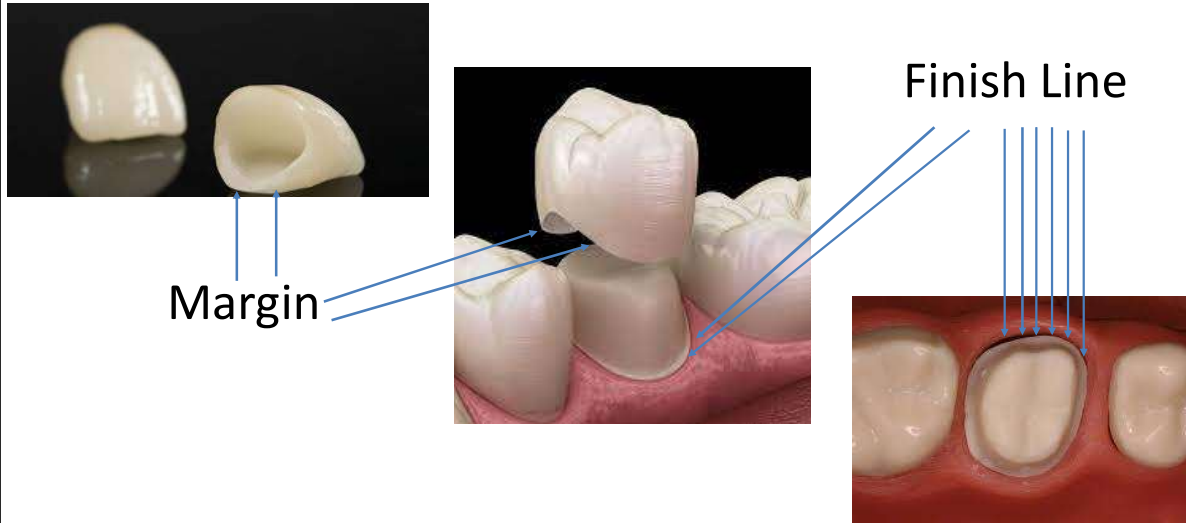
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Crown to root ratio is defined as the physical relationship between the portion of the tooth within alveolar bone compared with the portion not within the alveolar bone. This is determined with a good radiograph. We want a ratio that is less than one. This means there is more tooth in the bone than out. The more ideal relationship would be 2:3 as can be seen in tooth A in the picture. Tooth B represents a tooth with a compromised ratio that is more than one. This will give this tooth a poor prognosis for long term success.

Crown Margin and Tooth Finish Line



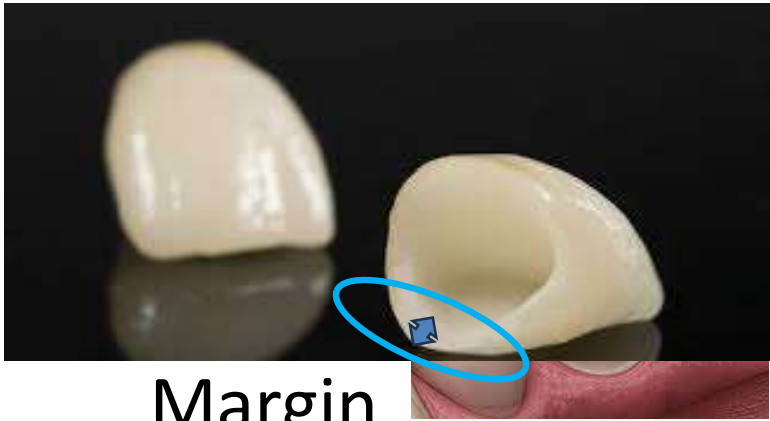
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A crown margin is the outer edge of a crown. This edge will interact with the boundary surface of the tooth preparation. This boundary surface is known as the finish line or finish curve.

Crown Margin and Tooth Finish Line



Margin

Finish Line



A crown margin is the outer edge of a crown. This edge will interact with the boundary surface of the tooth preparation. This boundary surface is known as the finish line or finish curve.

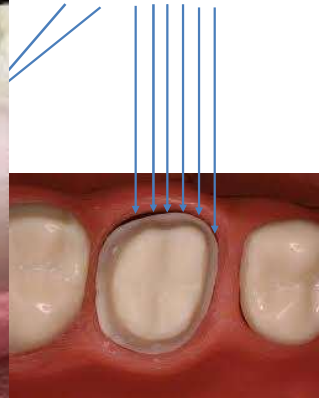
Crown Margin and Tooth Finish Line



Mar



Finish Line



10

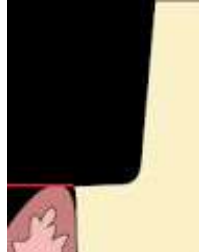
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A crown margin is the outer edge of a crown. This edge will interact with the boundary surface of the tooth preparation. This boundary surface is known as the finish line or finish curve.

Margin/Finish Line Placement



Sub-gingival



Crestal



Supra-gingival



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When preparing a crown there are three places that a crown margin can be placed on a tooth.

- 1) Supra-gingivally
- 2) At the gingival Crest
- 3) Sub-gingivally

Many factors will go into choosing where to place your margin. These can include decay location, margin of an old restoration or crown, material used, the emergence profile and patient's esthetics concerns. We will discuss this more in a future lecture.



Crown Preparation Steps

- Buccal and Lingual Axial Reduction
- Interproximal Reduction
- Occlusal Reduction
- Functional Cusp Reduction/Bevel
- Smoothing and Refining
- Final Evaluation/Assessment

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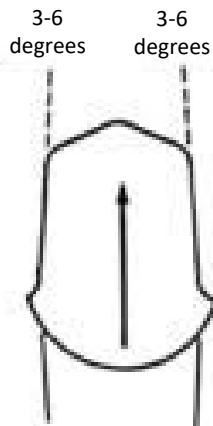
Now that you have decided to prepare a tooth for a crown and determined the position of your margin, it is time to start the preparation process. There are 6 steps to crown preparations. At CNU we have chosen to do it in this order. Other schools might prepare a tooth in a different order. The 6 steps are:

- 1) Buccal and Lingual Axial Reduction
- 2) Interproximal Reduction
- 3) Occlusal Reduction
- 4) Functional Cusp Bevel
- 5) Smoothing and refining
- 6) Final Evaluation and assessment

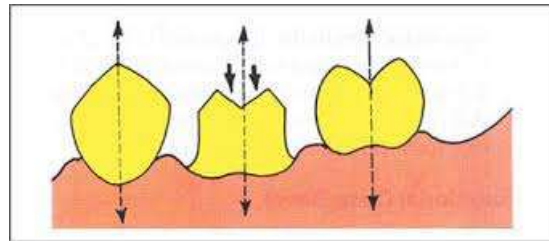
We will review each step of the process individually.

Taper & Line of Draw

Taper



Line of Draw



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Taper is defined as: The convergence of two opposing external walls of a tooth preparation as viewed in a given plane. An example of this would be the Mesial to Distal walls or Buccal to Lingual walls. Generally, we view the taper from the occlusal in a single plane which is known as the line of draw or path of insertion. Ideally, the taper for a crown is 6 degrees. If you hold your diamond bur in a constant angle you will be able to achieve a 6 degree taper. This is very hard to achieve. It has been shown that a taper of 10-12 degrees is acceptable.

Line of Draw or Pathway of Insertion is the path that a crown will follow to be seated onto a tooth. This is a consistent direction. On the right side of the slide the middle tooth has been prepared for a crown. You can see from the arrows that the pathway a crown will seat on this tooth is in line with the long axis of the tooth. Generally, this is what we want to achieve. There be times in your career you will have to deviate from this.

Taper/Line of Draw



Axial Reduction



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We will start our preparation with the reduction of the buccal and lingual axial wall. The axial wall is the vertical walls created on the buccal, lingual, mesial and distal. Each material will have a specific requirement for the amount axial reduction. For a porcelain crown the thickness or depth at the margin needs to be 1.0-1.5mm while a porcelain fused to metal (PFM) is 1.0-1.25mm and a gold crown is 0.5mm.

To start the axial reduction, you will want to make depth cuts or grooves on the Buccal and Lingual surfaces of the tooth. This can be done with your chamfer diamond, round bur or a carbide. The placement of the apical portion should be slightly above the location that you would like your final margin. This will allow you leeway to do your finishing and smoothing of the final margin.

The bur should be held in the long axis of the tooth to help generate the appropriate taper

Once you have made your depth grooves you will want to connect them giving you a continuous reduction on the B and L surfaces. Again, while you are connecting the depth grooves, hold your diamond in the long axis of the tooth. At this time, the preparation will appear rough and jagged. We will smooth this out later in the process

Video



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This is a video demonstrating Placing depth grooves. You will want to place your depth grooves a little occlusal of where you would like your final margin to be. This will give you some space for smoothing and correction.

Video



This is a video demonstrating the connecting of the depth grooves.

Interproximal Reduction



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Now that you have reduced the buccal and lingual you will want to connect them. There are 2 ways to do this. The first is to prepare in a brushing motion from the Buccal to the Lingual. The second technique is to prepare down from the occlusal like an interproximal filling. When preparing the interproximal portion of the tooth it is important to remember that the tooth is 1.5-2.0mm wider at the contact point than at the CEJ. Understanding this will help to prevent damaging the neighboring tooth during preparation.

Interproximal Reduction



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Interproximal Reduction



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Video



This is a video demonstrating axial reduction process interproximally from the Buccal.

Video

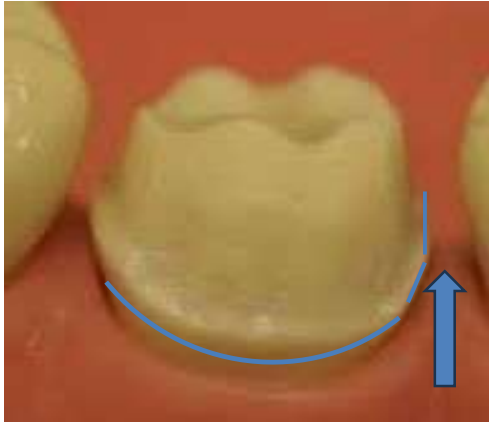


This is a video demonstrating axial reduction process interproximally from the occlusal.

Video

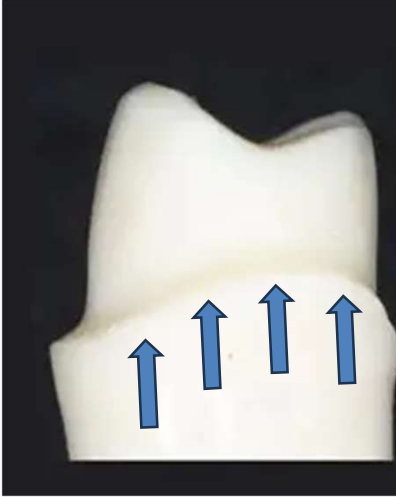


Margins

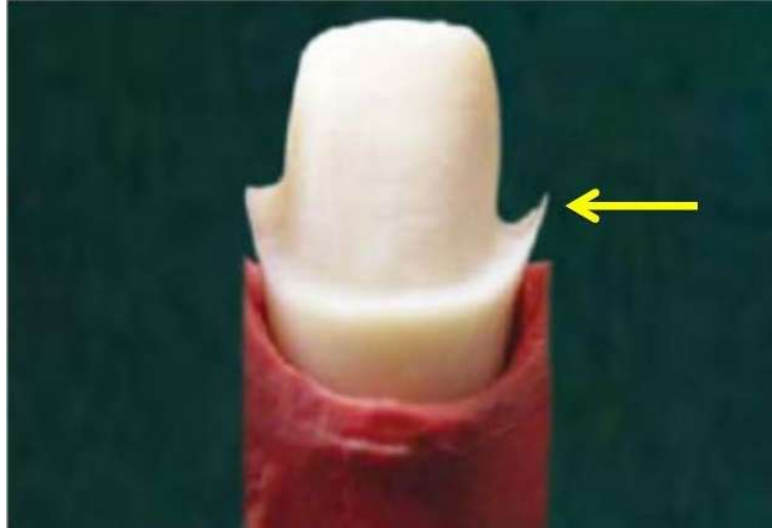


Now that we have done our basic or gross axial reduction, we need to discuss margin shape. The margin should follow the shape that the CEJ forms. This means that the margin will be higher interproximally than on the buccal and lingual surfaces. Additionally, the margin needs to be smooth. This means that there are minimal to no irregularities or steps. These things will reduce the adaptation of the crown to the tooth. A smooth margin will also make it easier to impress the tooth, perform laboratory steps, scan and mill.

Margins



“J” Shaped Margin



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The “J” shaped margin AKA: the ski jump, is caused by sinking a rounded diamond beyond the width of the cutting tip. This creates unsupported enamel that will need to be removed. This can be done with the end cutting diamond in your bur block. One way to prevent this is to use a diamond with a slightly different shape. We will discuss this on the next slide.

Diamonds

Diamonds: Round vs. Multi-Radius



The typical rounded diamond seen on the Left side is very good for making rounded internal walls but has a tendency to generate the “J” shaped wall. A more modern shaped diamond known as the Multi-radius end will still give you a rounded internal shape but be better at preventing the “J” shape. Our main diamond in your bur block is a Multi-Radius style of bur.

“J” Shaped Margin Removal

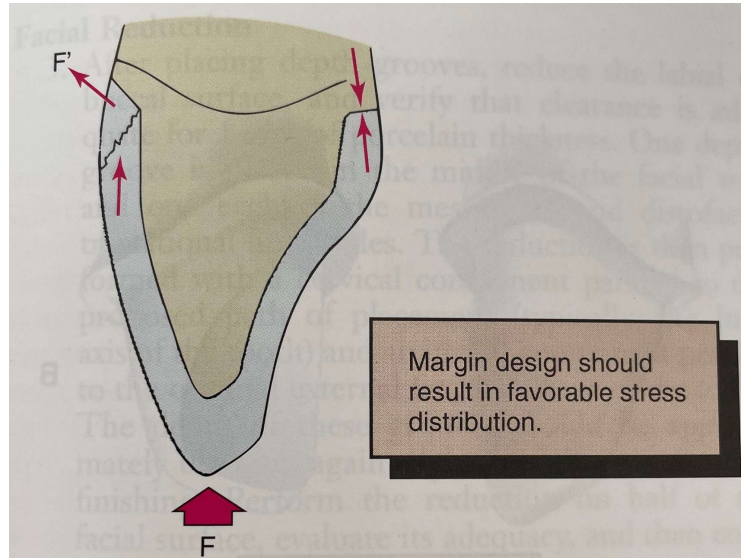


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Final Margin Shape



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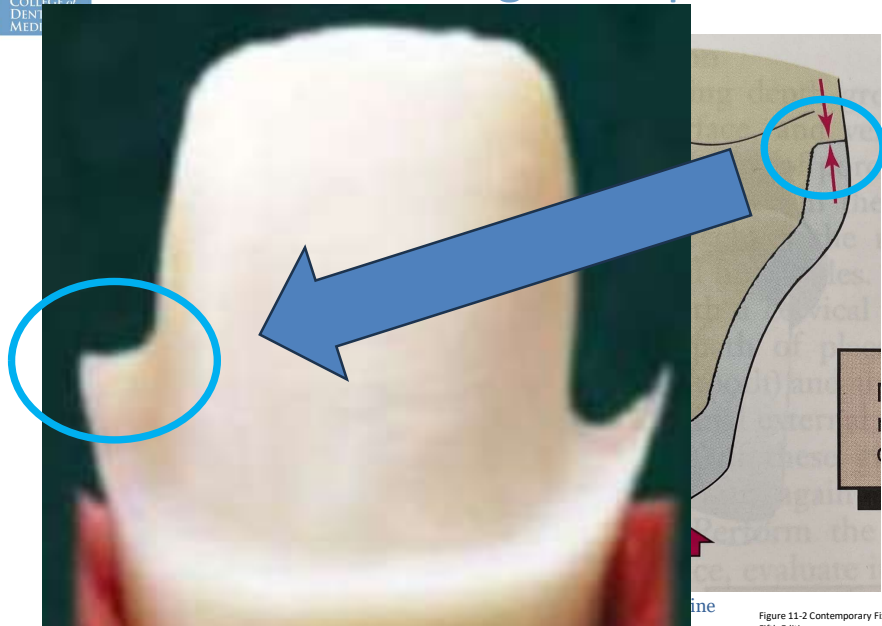
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Figure 11-2 Contemporary Fixed Prosthodontics
Fifth Edition

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Your final margin shape should be a butt joint with a 90 degree exit angle. As can be seen in the picture in the slide the lingual margin is a flat butt joint reducing the stress on the material. The buccal margin is sloped causing stress in the material which will eventually cause fracture and failure. The design on the buccal is for a PFM that has metal support underneath.

Final Margin Shape



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Figure 11-2 Contemporary Fixed Prosthodontics
Fifth Edition

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Occlusal Reduction



Half of the occlusal reduction is performed; the other half is maintained for reference purposes.

Pictures from Contemporary Fixed Prosthodontics 6th Edition

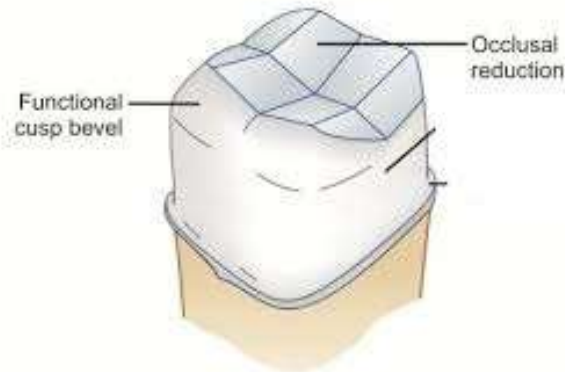
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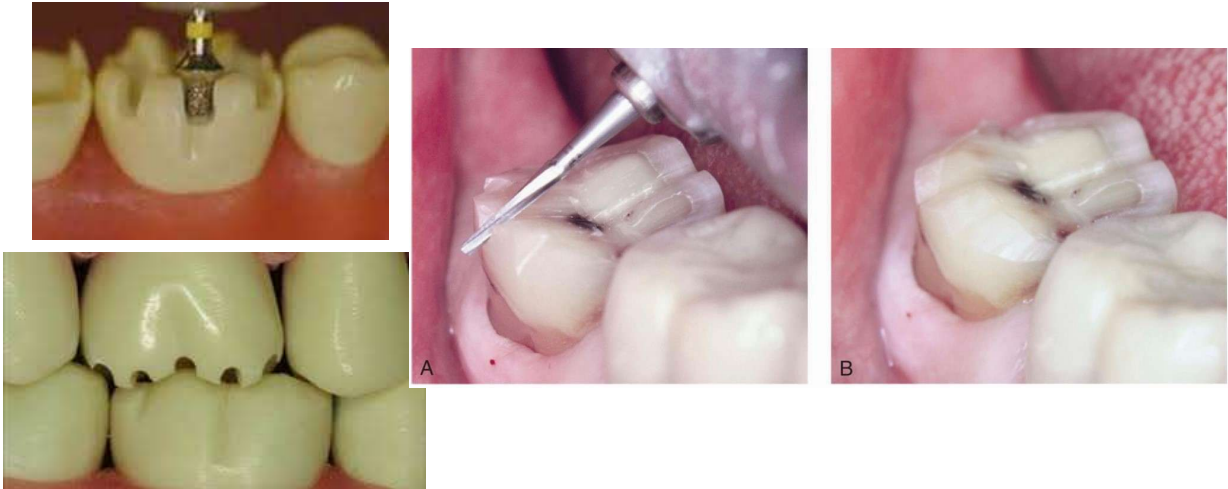
Now it is time to move onto the reduction of the occlusal surface. Unlike a crownlay where we are not paying much attention to the planes of the tooth, for a full coverage restoration we need to take this into account. When preparing a full coverage crown, we want to follow the planes of the tooth. If we only follow the planes on the occlusal table, we will not have the proper clearance with the opposing tooth. Occlusal clearance is the amount of reduction achieved during tooth preparation to provide for the adequate thickness of restoration material. To achieve the proper clearance, we need to have a functional cusp bevel. Also, along on the occlusal surface we want to have morphological reduction to preserve tooth structure.

Morphological Reduction



Morphological reduction is the process of reducing the occlusion while following the differing planes of the occlusal table. As you can see in the picture there are distinct planes going down from the triangular ridges into the grooves and fossa. This will give you lab technician the proper clearance while also preserving tooth structure.

Occlusal Reduction



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So how do you do the occlusal reduction. As with the axial reduction you want to make depth grooves in the occlusal table. This can be done with your chamfer diamond or the depth cutting bur in your bur block. An example of this bur is seen on the upper lefthand side picture. Once your depth grooves have been cut you will want to connect them with your chamfer diamond following the planes of the tooth. The rough reduction is seen on the righthand side picture.

Proper Reduction



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The black lines indicate either a high point or low point on the occlusal table. The Triangular ridges and oblique ridge on a maxillary molar will be your high points and the grooves will be a low point.

Depth Grooves



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Depth grooves have been placed on the buccal of the occlusal table. The depth grooves are placed at the same plane angle that the buccal cusp create.

Occlusal Planes on the Buccal



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The buccal occlusal reduction has been completed. Notice that the former triangular ridges are the high point of a “V” or hill on the occlusal table and everything slopes down from this point.

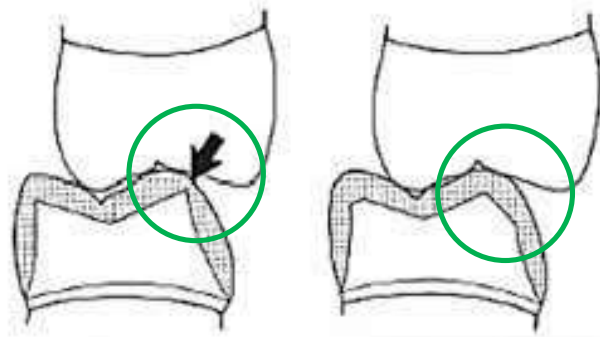
Prepped Occlusal Table



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The lingual occlusal reduction has now been completed, again with high and low points. The final appearance should mimic the anatomy of the occlusal table.

Functional Cusp Reduction (AKA Functional Cusp Bevel)



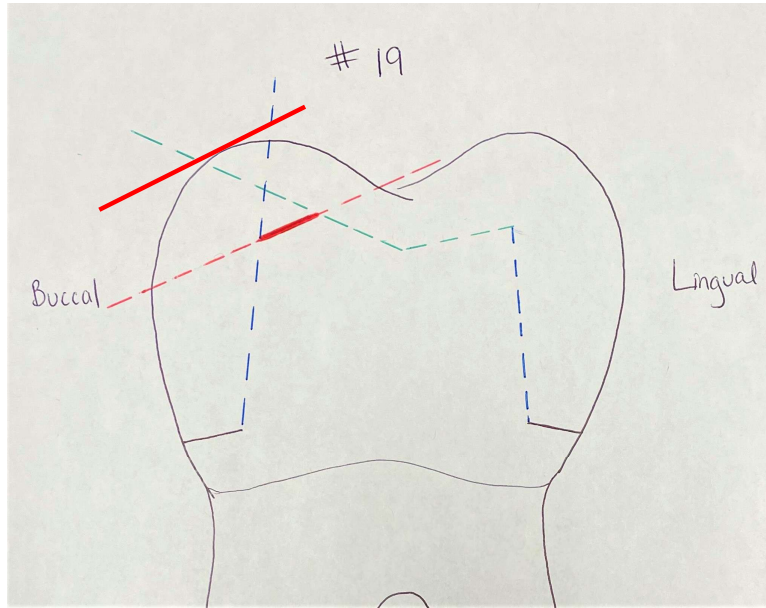
The maxillary and mandibular teeth have a functional cusp that will contact the marginal ridge or central fossa of the opposing tooth. This is the Lingual cusp for the maxilla and Buccal cusp for the mandible. If we do not place a functional cusp bevel there will not be enough clearance in this area for the restorative material.

Functional Cusp Reduction

- This is **NOT** a bevel that you would do on a Composite restoration
- **NOT** a 0.5-1.0mm bevel
- It should be **Large** and a **3rd plane** in your occlusal reduction
- It should follow the angulation of the **lingual-occlusal** surface for the maxilla or the **buccal-occlusal** surface on the mandible

You will now have to disassociate the word bevel when it comes to operative and fixed. A bevel in operative is a small change of direction along the cavosurface margin. The functional cusp bevel is a portion of the occlusal reduction in a posterior tooth. It is not 0.5-1.0mm in length. It is not flat but follows the angulation that the functional area of the tooth. This is the buccal-occlusal surface on a mandibular tooth and the lingual-occlusal surface on a maxillary tooth.

Functional Cusp Reduction



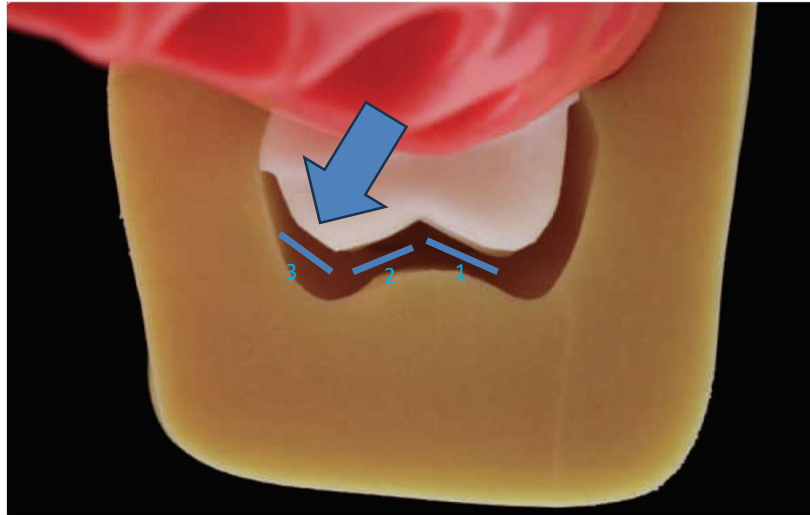
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This is a view from the interproximal of #19. The axial wall is represented in blue. The occlusal reduction of the occlusal table is represented in green. The Functional Cusp Bevel is represented in red. This produces a third plane that connects the axial wall to the occlusal table

Functional Cusp Reduction/Bevel

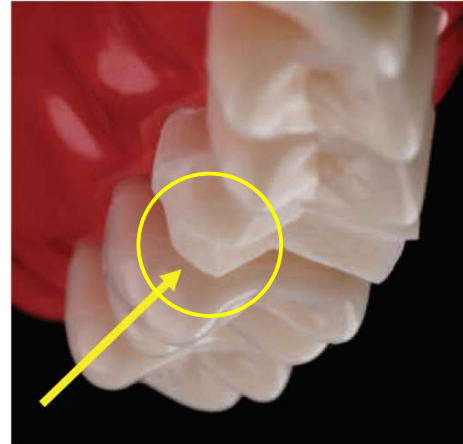


As you can see in this cross-sectional view of the preparation and putty stent, there 3 distinct planes of occlusion on the tooth. The functional cusp “bevel” forms that third plane.

Functional Cusp Reduction/Bevel



NO!



Appropriate

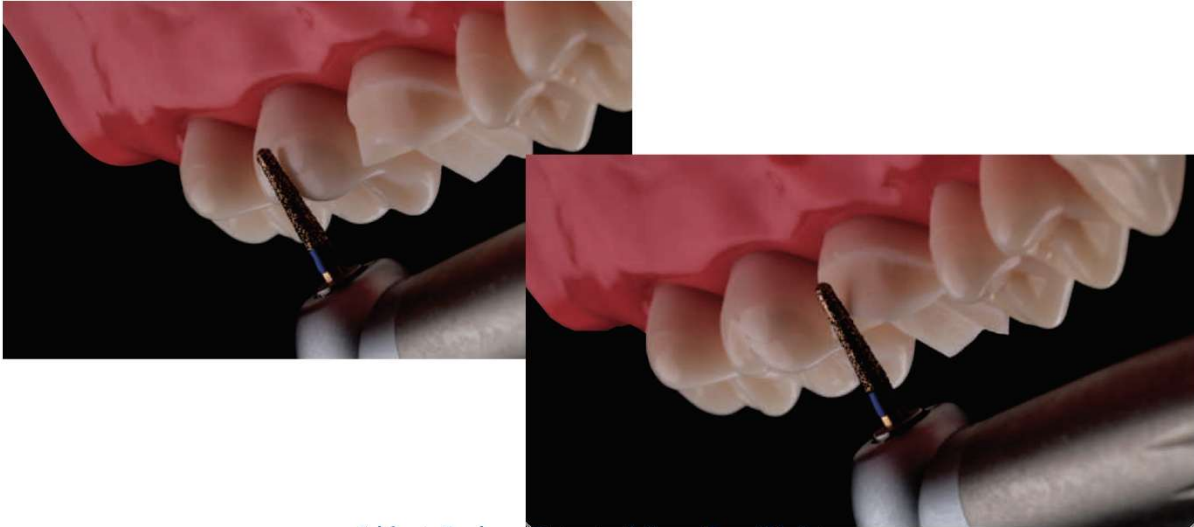
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When placing the functional cusp bevel you can see this is not a small composite bevel. It is a third plane of occlusal reduction. The portion of the tooth that is reduced is the area that is outside of the occlusal table but not the Buccal or Lingual surface. The size is only slightly smaller than the 2 other planes on the occlusal table.

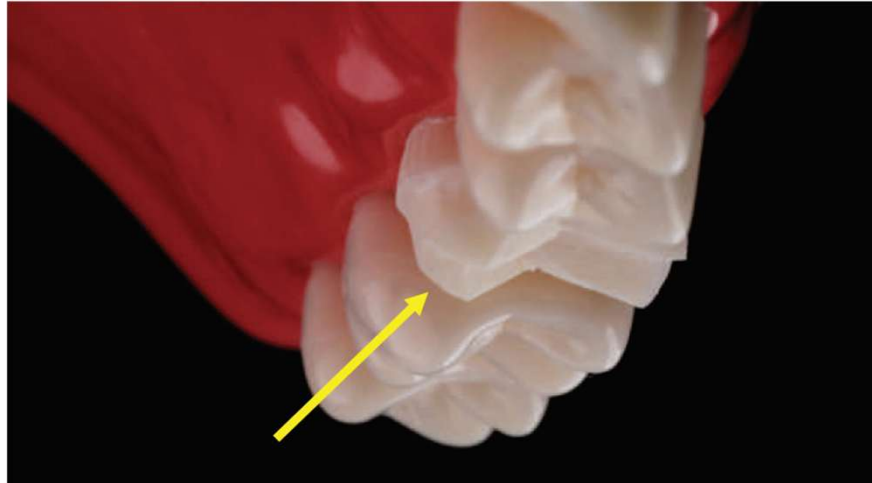
Functional Cusp Reduction/Bevel



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The Functional Cusp Bevel is **NOT** a composite bevel. This means that it is **NOT** a 0.5mm bevel. It is occlusal reduction of the functional cusp forming a third plane of the occlusal table. On Mandibular teeth this will be placed on the buccal surface of the buccal cusp. For Maxillary teeth, the bevel will be placed on the lingual surface of the lingual cusps. You will need to angulate your bur to follow the angulation of the tooth in that area.

Functional Cusp Reduction/Bevel



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As you can see the lingual of the lingual cusps form a distinct plane on the lingual surface.

Finished Occlusal Reduction/Bevel



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This shows a completed occlusal reduction for a maxillary molar. Notice that the planes prepared mimic the anatomy and you should be able to identify the type of tooth following your reduction.

Video



This video demonstrates the placement of depth grooves in the occlusal table.

Video



In this video you will see how to connect the depth grooves to create the differing planes of the occlusal surface.

Finishing the Preparation



Finishing the Preparation



Finishing the Preparation



Finished Preparation



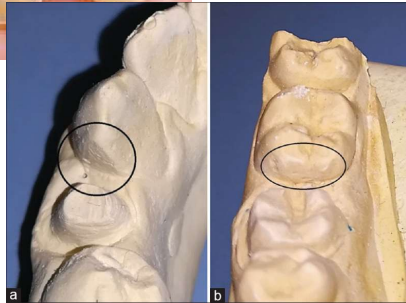
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This is an example preparation from all different directions. Please note all of the different planes that exist in the preparation. Also pay attention to the fact that the triangular ridges line up with each other in a B-L direction and not facing the central pit. Additionally, note that the functional cusp bevel is similar in length as a portion of the buccal or lingual slopes of the occlusal table.

Preventing Damage



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As you can see, we have now removed a lot of tooth structure. In this process it is possible to cause damage to the tooth and surrounding structures. Some common sites are the adjacent tooth, the local soft tissue structures like the gums, tongue or cheek and the pulp of the tooth. Most of the time a rubber dam is not used to prepare a tooth for a crown. The dentist and assistant need to be paying attention to the patient's tongue and cheeks to ensure they do not cut them.

Adjacent Tooth Damage



As with Approximal Cavity Preparations, it is possible to damage the adjacent tooth. Most of the time this is done when doing your interproximal axial reduction. It is really important to remember that teeth are generally 1.5-2.0mm wider at the interproximal contact point than they are at the CEJ. So, if you margin will be below the contact you will need to take this into account when choosing your starting point for the axial reduction.

Preparation Evaluation



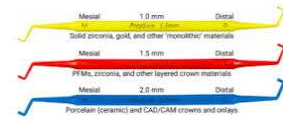
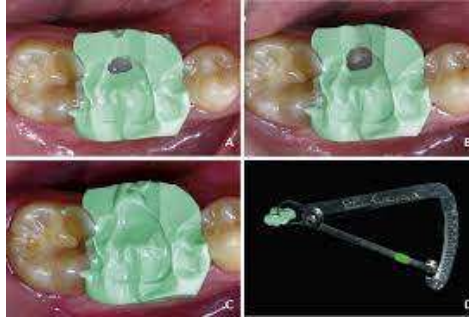
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When you feel like your preparation is complete, you want to view the preparation from the occlusal and look at the prep to see if you can see all of the axial walls. You are looking for undercuts, or an irregularity of the prepared wall that would prevent seating or removal of a restoration or prevent a lab technician from waxing a crown. When you see an undercut, you will want to take your chamfer diamond and smooth that area. Next, you will want to evaluate your margin and make sure it is smooth and not jagged or have irregularities at the finish line. This can be checked visually or with the end of an explorer. If the margin is rough, you will want to smooth it with your end cutting diamond or your chamfer diamond without water and with a very gentle touch. Next, you want to make sure that your axial depth is correct circumferentially. Finally, you want to make sure that your finish line follows the contours of the gingival tissues not only on the buccal and lingual surfaces but on the mesial and distal surfaces. This will create an interproximal margin that higher occlusally than the buccal or lingual.

Preparation Evaluation: Occlusal Reduction



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Once you are satisfied with your overall preparation it is vital to check for the occlusal clearance. Just because you made proper depth grooves and connected them, there is no guarantee that your clearance will be correct. You can check this in many ways. First, you can do this visually. This is the least accurate since it is hard to visualize the lingual cusps to make sure you have the proper clearance. Next, you can use prep check devices. These come in the form of instruments or bite tabs that will tell you your clearance. For an instrument, the tip should freely move between the teeth and the bite check tabs should easily slide out when the patient is in MIP. Lastly, you can use rigid bite registration material and have the patient bite down in MIP and let the material set. Then use your calipers to determine the thickness of the material to see if you have enough clearance. This is demonstrated by the image in the middle of the slide.

Video



This Video shows the evaluation of your preparation.

Material Choices



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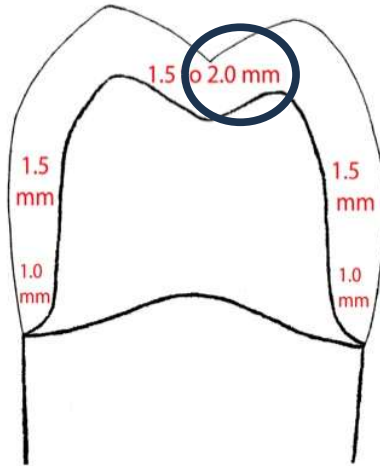
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Today there are many choices for porcelain posterior crowns. These range from Emax and zirconia to porcelain fused to zirconia (LAVA by 3M) to porcelain fused to metal or gold. Each of these materials will have their positives and negatives. Crowns that have porcelain on their occlusal surface will be more abrasive than metal or natural teeth. This will cause more wear to the opposing tooth structure. Additionally, the preparation design and requirements for space are different. An all-ceramic crown will have the advantage of higher esthetics and a more natural appearance, but they do not have any substructure metal to resist the occlusal forces. Additionally, all-ceramic crowns require greater occlusal and axial reduction and they have a tendency to be more brittle. This has improved with the introduction of Emax and all zirconia crowns.

A crown with a metal substructure will have greater support to withstand the occlusal forces but they have challenges as well. First, they are not as esthetic due to the graying effect of the metal. Secondly, the porcelain has to be fused to the metal and this can be a failure point in the crown. Lastly, the porcelain will need to be supported underneath by the metal. If you have too much porcelain without supporting metal that area can also be a location of failure. With PFM crowns you can have all metal margins, partial metal margins (only on the lingual) and all porcelain margins. Additionally, your occlusal can have all or part metal. The metal will do better with the occlusion and wear the opposing teeth less but it is not esthetic.

Reduction Requirements

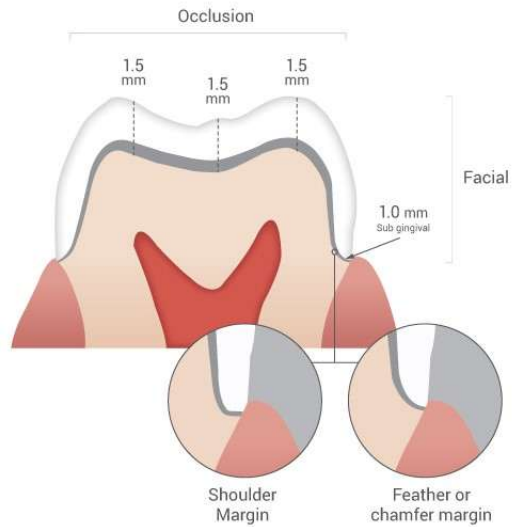
All Porcelain Crowns



The preparation guidelines for an all porcelain crown are displayed in the picture in the slide. Ideally, the occlusal reduction should be 2mm for Emax and other combination type materials. All zirconia crowns can have an occlusal reduction amount as low as 1 mm but this is testing the limits on the strength of the material and it will be more prone to fracture. The depth at the margin should be at least 1mm, which will generate an axial reduction of at least 1.5 mm on the mid wall. All areas of the preparation should be rounded. These preparation parameters can also be used for Porcelain Fused to Zirconia (PFZ). An example of this would be a Lava crown by 3M.

Reduction Requirements

PFM/PFZ crowns with
Porcelain Occlusion



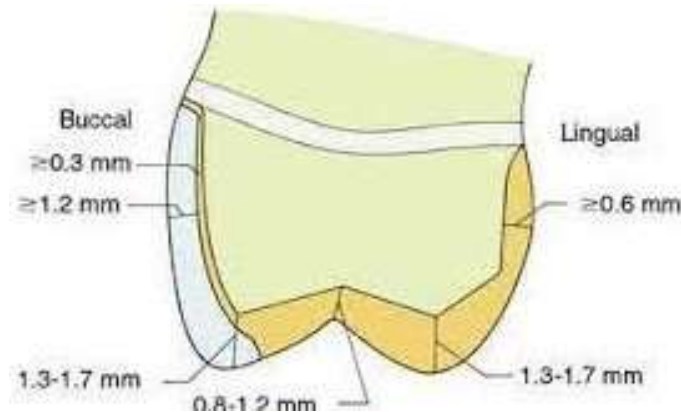
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For a PFM that will have all porcelain occlusal the reduction parameters are shown in the slide. It is very important to have the proper reduction on the Buccal and Lingual to give the proper material thickness without having the crown be overcontoured.

Reduction Requirements



PFM with Metal Occlusion and Lingual Margin

When doing a PFM with a metal occlusal and lingual the reduction requirements change. For the occlusal portion of the prep 1.5 mm of reduction will satisfy both metal and porcelain requirements. On the Buccal, 1-1.5 mm will be needed for the proper thickness of the metal coping and porcelain. The lingual will need 0.5-0.6mm of reduction .